

Passive Device Matching

- Absolute value is not important ($\pm 20\%$)
- Ratio is important, e.g. $A_v = -R_1/R_2$ for inverting amplifier
- The larger the area, the better the matching (Why?)

Hints: $R = \rho \frac{\text{length}}{\text{thickness} \times \text{width}}$ $C = \frac{\epsilon_{ox}}{t_{ox}} \times A$

- to make R1 and R2 track well with each other, use the R1 and R2 = 10 x R1
- Replace the equation with the $x + \Delta x$
 - then factor out x will get $\Delta x / x$ small

Handwritten notes on a whiteboard showing the derivation of the matching equation. The notes include the formulas for resistance R and capacitance C , and a derivation of the matching equation using the binomial approximation. The final equation is $C = \frac{\epsilon_{ox}}{t_{ox}} \times A \times \left(1 + \frac{\Delta l}{l} + \frac{\Delta w}{w}\right)$.

- ■ approximations
- in PDK there is pelgrom coefficient
 - <https://www.eecg.utoronto.ca/~ali/papers/mag-win-15-process-variation.pdf>
 - <https://www.edaboard.com/threads/mismatch-pelgrom-coefficient.376720/>

Example

Example : 81k Ω versus 44k Ω ?

$$1\text{k}\Omega \times 81 = 81\text{k}\Omega$$

$$1\text{k}\Omega \times 44 = 44\text{k}\Omega$$

Is this the best approach?

- using all 1k is best matching ? no

Example : 81kΩ versus 44kΩ

$$40\text{k}\Omega \times 2 + 1\text{k}\Omega \times 1 = 81\text{k}\Omega$$

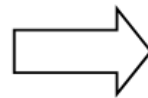
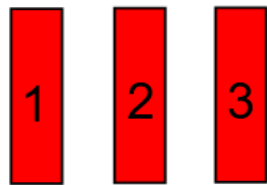
$$40\text{k}\Omega \times 1 + 1\text{k}\Omega \times 4 = 44\text{k}\Omega$$

Moral of the story : Think Big

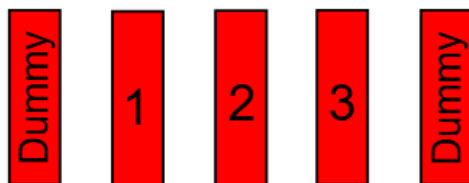


- find the largest common factor
 - because the error for matching comes from the big value
 - 1% mismatch in 40k is 400
 - 5% mismatch in 1k is 50
- so should match the dominant 40k
- mismatch in the 1k doesn't matter much

Matching with Dummy



1, 2 and 3 are not matching



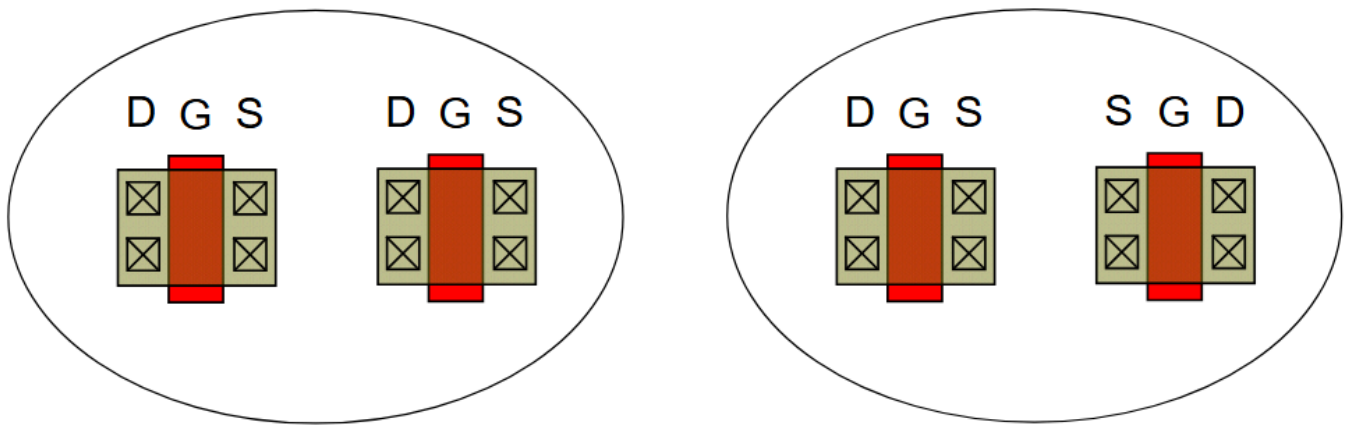
1, 2 and 3 are matching

- Try to make each unit see the same surroundings

- from manufacturing
 - the implantation process might be slanted, so with shadowing effect, 1 and 2/3 is different
 - then, 3 might reflect shit back to 2, so 2 and 3 is not the same
- by using dummy, can make the manufacturing effect the same,
 - the effect is by proximity, so one dummy is enough

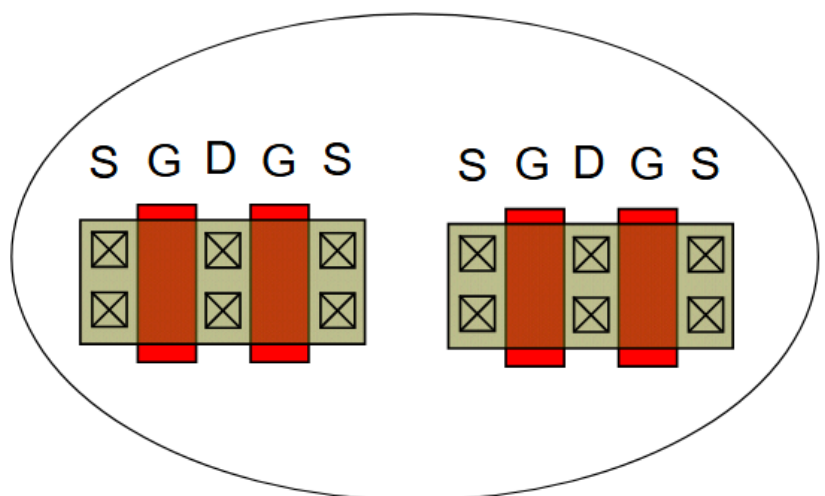
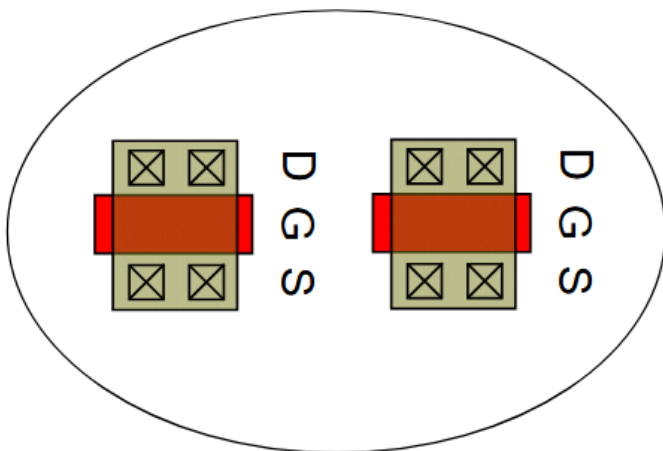
- but if the feature size is small, need more dummies

Active Device Matching



• Which layout matches better?

- left is better
- fabrication is by layer, any shift in layer will affect the left two MOS in the same way, but will make the right two MOS in a very different way (one is D bigger, the other is S bigger)



- top left is a rotated of the good layout from previous slide
- with advanced process of strained technology, device orientation is important, so cannot anyhow rotate
 - based on the direction of expansion, the other dimension shrinks
 - the expanded direction increase in electron mobility, because the atoms are further apart
 - so only one direction is good for the transistor

- This strained technology is to increase the electron mobility, so they can move faster and mosfet can switch faster
 - it is the stretching of the wafer itself
 - https://en.wikipedia.org/wiki/Strained_silicon
- Another way is to make the device even number fingers so the layer shift will not affect it